Quant Edge 1.0 | Final Challenge

Capital Preservation First: A Tail-Risk-Aware Portfolio for the Serendib University Endowment Fund

Our recommendation is to accept the mandate as proposed, with the portfolio and risk limits set out in this report. The portfolio holds 61% in AGG, 19% in GLD, 15% in XLK and 5% in SPY, with EFA, EEM and XLE at zero. It is rebalanced monthly, long only and unleveraged, and every position comes from the prescribed investable universe.

Team : Quantix
Tracking Code: SAIFA-2026-9F6E94FE
Data : 3rd January 2005 to 27th October 2026 (5,489 trading days, 7 ETFs)

1. Executive summary

This report presents the quantitative case for a USD 10 million mandate from a conservative university endowment, whose Investment Committee holds that "a good investment decision is not the one with the highest expected return; it is the one whose downside you have actually measured, tested, and can defend." An end-to-end Python pipeline was developed covering portfolio construction, performance KPIs, three Value-at-Risk methodologies, Expected Shortfall, Extreme Value Theory, out-of-sample back-testing, stress testing and risk budgeting. All figures presented herein are derived from a single cleaned price dataset and are reproduced by the accompanying analysis script.

Seven findings that drive the recommendation

- **Construction.** Maximum-Sharpe optimisation without limits produces a 46% gold / 50% technology portfolio with 33% drawdown, which fails the mandate. Minimum variance is 92% bonds and earns only 3.7%, below the return objective. We therefore maximise Sharpe inside an explicit Investment Policy Statement (IPS) with a 7% volatility limit and asset caps, giving the recommended weights.
- **Performance.** CAGR 7.0%, Sharpe 0.67, Sortino 0.95, Calmar 0.39, market beta 0.22, Jensen alpha 2.4%. Down-market capture is 19% against 37% up-market capture (monthly, vs SPY).
- **Three VaR methods disagree where it matters.** At 99% the normal model gives 0.97% VaR versus 1.13% historical, and understates Expected Shortfall by 31% (1.11% vs 1.62%). Portfolio returns have excess kurtosis of 17.6, so normality is rejected decisively.
- **EVT quantifies the far tail.** A generalised Pareto fit (threshold at the 91% loss quantile, shape 0.20) gives a 99.9% one-day ES of 2.99%, about 2.1 times the normal estimate of 1.41%.
- **Backtesting selects the operating model.** Over 4,738 out-of-sample days, EWMA-filtered historical simulation (FHS) recorded 50 99% exceptions against 47 expected (Kupiec $p = 0.705$) and is in the Basel green zone today. The normal model recorded 82 and is rejected ($p < 0.001$).
- **Stress tests show where the protection fails.** GFC -5.3%, COVID -7.6%, 2022 -17.6%; our hypothetical real-rate shock (+200bp) costs -14.7% (USD 1.47m).
- **Risk is concentrated and regime dependent.** Gold and technology supply 65% of volatility; in turbulent regimes one-day 99% ES rises to 2.42% from 1.25% in calm regimes.

Metric (2005-2026, monthly rebalanced)	Recommended	60/40 SPY/AGG	SPY
Compound annual return (CAGR)	7.0%	7.9%	10.9%
Annualized volatility	6.8%	11.3%	18.9%
Sharpe ratio (rf = 2.5%)	0.67	0.51	0.51
Maximum drawdown	-17.9%	-35.8%	-55.2%
One-day 99% VaR / ES (historical, % of NAV)	1.13% / 1.62%	n/a	n/a
One-day 99% EVT VaR / ES	1.15% / 1.62%	n/a	n/a
GFC / COVID / 2022 stress return	-5.3% / -7.6% / -17.6%	-27.0% / -20.6% / -20.6%	-46.4% / -33.4% / -23.8%

Recommendation

Accept the mandate with the portfolio in Section 5 and the limit framework in Section 12: an ES-based daily risk limit, a drawdown-triggered review at -12%, and a standing watch on the stock-bond correlation. We recommend against both the minimum-variance and maximum-Sharpe alternatives, for the reasons quantified in Sections 5 and 6.

2. Mandate interpretation and Investment Policy Statement

The brief leaves horizon, return objective and risk constraints to the manager. We set them from the client's profile (a perpetual endowment funding scholarships, research and faculty development, with strong emphasis on capital preservation) and then tested every one of them against the data.

IPS element	Our setting	Justification
Investment horizon	Long term, 10+ years; monthly rebalancing	Perpetual endowment; spending needs are recurring, so forced selling in a drawdown is the main threat.
Return objective	At least 6.5% p.a. nominal	Approximately 4% annual spending plus 2.5% inflation. Minimum-variance (3.7%) fails it; 60/40 (7.9%) meets it only by breaching risk limits.
Volatility limit	At most 7% p.a. (ex-ante)	Roughly 60% of the volatility of a 60/40 portfolio, consistent with a conservative profile.
Tail-risk limit	One-day 99% ES at most 2.0% of NAV (USD 200k)	Expected Shortfall is the committee's loss-in-the-tail measure; cross-checked with EVT.
Drawdown tolerance	At most 20% peak to trough	Scholarship funding is protected if the portfolio never falls below four-fifths of its high-water mark.
Stress-loss tolerance	At most 20% in any scenario (USD 2.0m)	Applies to the three historical crises and our hypothetical shocks.
Concentration	Asset caps: SPY 25, EFA 10, EEM 5, AGG 40-60, GLD 20, XLE 5, XLK 15 (%); no asset above 35% of risk	Keeps a bond core (at least 40%), limits the highest-volatility sleeves (EEM 27.6%, XLE 29.7% annual vol) and forces diversification.
Implementation	Long-only, unleveraged, ETFs only; risk-free rate 2.5%	Mandate and universe constraints. The 2.5% constant is a proxy for the 3-month T-bill (see limitations).

3. Data Sources and Cleaning Discipline

All analysis uses one file, price_data.xlsx, built from Yahoo Finance adjusted closes through the yfinance package (the approved primary source). It contains 5,489 daily prices from 2005-01-03 to 2026-10-27 for the seven prescribed ETFs (SPY, EFA, EEM, AGG, GLD, XLE, XLK) and 5,488 daily returns. The cleaning steps required by the brief were applied in code, not by hand:

- **Adjusted closes** are used throughout, so dividends and distributions do not distort returns (important for AGG, whose return is mostly income).
- **Calendar alignment.** All seven ETFs are listed on US exchanges and share one trading calendar, so no cross-exchange alignment loss occurred: zero missing prices, zero forward-fills, zero dropped dates. International exposure (EFA, EEM) is therefore measured in USD at US close; the overnight timing gap between local markets and the US close is a limitation, not a cleaning step.
- **Return conventions.** Simple returns are used for portfolio aggregation, VaR, ES and EVT, because a portfolio return is the weighted sum of asset simple returns. Log returns are used for the stationarity tests and descriptive statistics.
- **Flag, do not delete.** Three single-day moves were flagged as implausible-jump candidates: EEM +22.8% (13 Oct 2008), EEM +20.9% (28 Oct 2008) and XLE -20.1% (9 Mar 2020). Each coincides with a documented market event (post-Lehman rebounds and the oil-price collapse) so all three were retained. A scan for returns beyond six standard deviations finds between 6 (GLD) and 13 observations per asset, which is the fat-tail signature we model rather than remove. No asset has a run of three or more identical consecutive prices (stale-price check).
- **Stationarity.** Augmented Dickey-Fuller tests on each asset's log returns (lag chosen by AIC) give statistics from -75.2 to -14.4, with the largest p-value 8e-27: the unit-root null is rejected for every

series, so returns are suitable for modelling. The portfolio return series also rejects it (ADF -16.5). Stationarity of the mean does not imply constant variance; see Section 4.

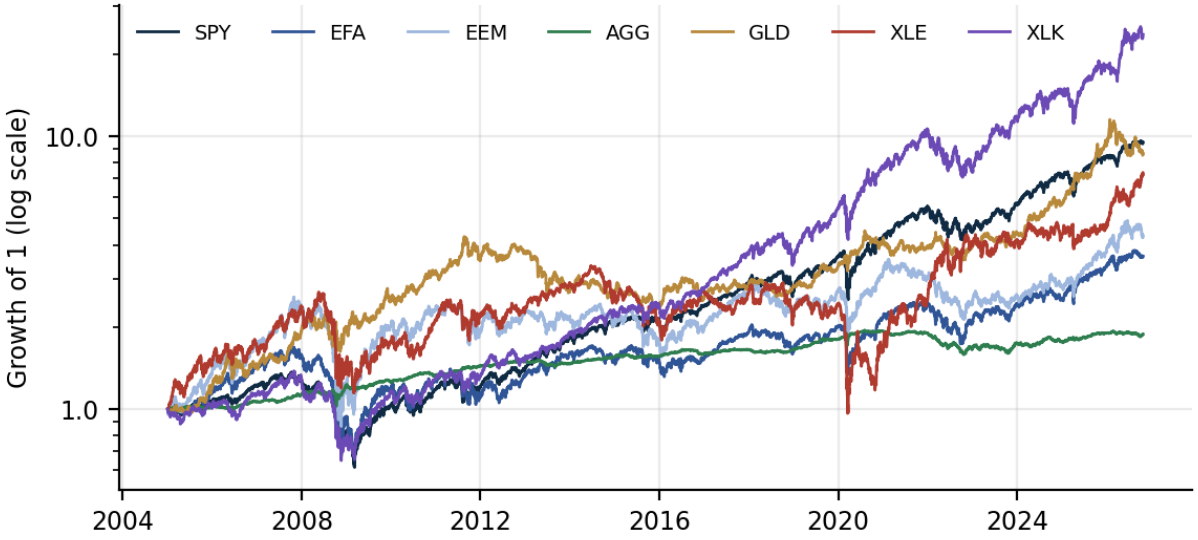


Figure 1. Adjusted-close price paths, normalised to 1 on 3 Jan 2005 (log scale). Technology and gold compounded fastest; AGG shows the stability that anchors the portfolio.

Ticker	Role	Ann. mean	Ann. vol	Skew	Ex. kurtosis	Worst day	Date
SPY	US large-cap	12.1%	18.9%	-0.00	15.5	-10.9%	2020-03-16
EFA	Intl developed	8.1%	20.9%	-0.04	14.3	-11.2%	2008-09-29
EEM	Emerging mkts	10.5%	27.6%	0.50	17.7	-16.2%	2008-10-15
AGG	US IG bonds	3.0%	5.2%	-1.84	50.5	-6.8%	2008-10-10
GLD	Gold	11.6%	18.3%	-0.31	6.4	-10.3%	2026-01-30
XLE	Energy	13.5%	29.7%	-0.32	11.6	-20.1%	2020-03-09
XLK	Technology	17.1%	22.6%	0.08	9.7	-13.8%	2020-03-16

Table 1. Descriptive statistics of daily simple returns, full sample (annualised with 252 days). Jarque-Bera rejects normality for every asset ($p < 0.001$).

4. Statistical properties and dependence

Three properties of the data determine which risk models are defensible. **First, fat tails:** excess kurtosis ranges from 6.4 (GLD) to 50.5 (AGG), against zero for a normal distribution. AGG, the supposed safe asset, has skewness of -1.84: its rare moves are large and negative (a 6.8% one-day loss in October 2008). **Second, volatility clustering:** a Ljung-Box test on the squared portfolio returns rejects independence at every lag we tried ($p < 0.001$), so yesterday's volatility predicts today's; an unconditional model will be too complacent after a shock and too cautious after a calm spell. **Third, time-varying dependence,** shown below.

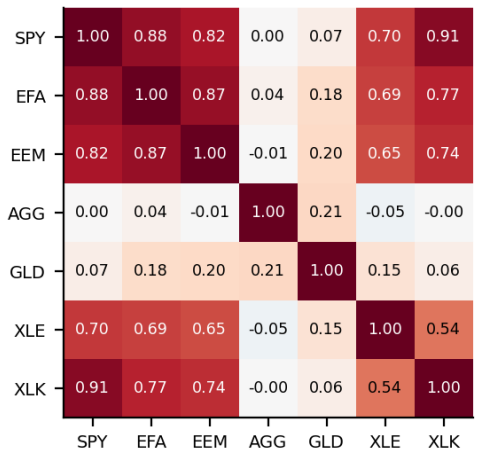


Figure 2. Daily-return correlations, full sample.

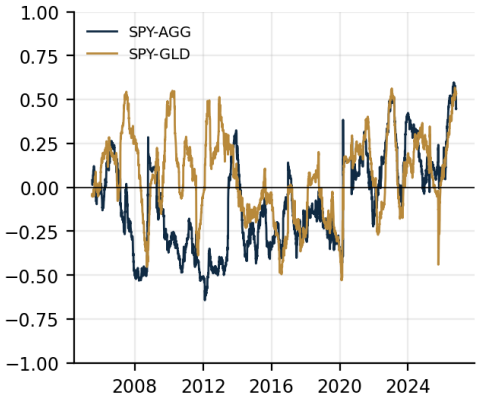


Figure 3. 126-day rolling correlation of SPY with AGG and GLD.

What the dependence structure tells us

- **The four equity-like ETFs are one risk factor.** SPY, EFA, EEM and XLK correlate between 0.74 and 0.91; adding EFA or EEM to SPY adds volatility and lower realised return (8.1% and 10.5% against 12.1%) without a diversification benefit. This is why the optimiser assigns them zero weight.
- **AGG and GLD are the true diversifiers.** AGG's correlation with SPY is 0.00 and GLD's is 0.07 over the full sample. Their mutual correlation is 0.21, so combining them creates a low-correlation core. XLE is the partial exception: 0.70 with SPY in aggregate, yet it rose +45% during the 2022 sell-off.
- **Diversification is not constant.** The 126-day SPY-AGG correlation has ranged from -0.64 (Feb 2012) to 0.60 (Sep 2026) and averages -0.09. It is +0.45 at the end of the sample, the regime in which stocks and bonds fall together. Any claim that bonds offset equity losses is conditional on this correlation, which we monitor in Section 12.

Portfolio-level return distribution

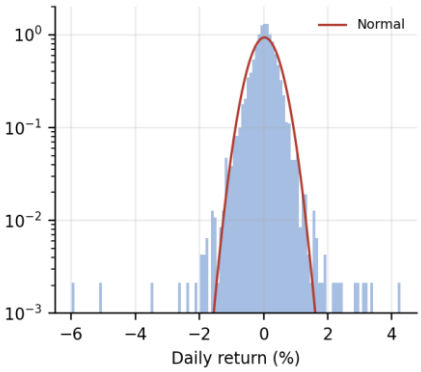


Figure 4. Recommended-portfolio return density against a normal curve (log scale).

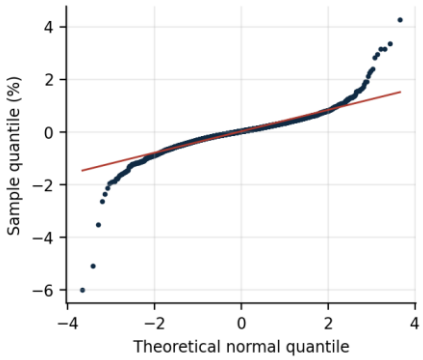


Figure 5. Normal Q-Q plot: both tails lie far outside the line.

The recommended portfolio's daily returns have skewness -0.57, excess kurtosis 17.6 (Jarque-Bera $p < 0.001$) and a fitted Student-t with 3.6 degrees of freedom, a tail so heavy that the fitted fourth moment is infinite. The worst day (6.0% loss on 2008-10-10) is 14.1 standard deviations from the mean, an event a normal model assigns a probability of effectively zero. These facts justify historical, Student-t and EVT measures alongside the normal benchmark, and conditioning on current volatility.

5. Layer 1: Portfolio construction (Markowitz and CAPM)

We start from Markowitz mean-variance theory using annualised sample means and covariances (252 trading days) and long-only weights summing to one. We estimate the CAPM against SPY as the market proxy with a 2.5% risk-free rate. CAPM clarifies what the optimiser is exploiting: it prices an asset's expected return from beta alone, so any gap between realised mean and CAPM-implied return is a (backward-looking) alpha.

Ticker	Beta vs SPY	CAPM expected return	Realised mean	Jensen alpha
SPY	1.00	12.1%	12.1%	+0.0%
EFA	0.98	11.9%	8.1%	-3.8%
EEM	1.20	14.0%	10.5%	-3.5%
AGG	0.00	2.5%	3.0%	+0.6%
GLD	0.07	3.1%	11.6%	+8.5%
XLE	1.11	13.2%	13.5%	+0.4%
XLK	1.09	13.0%	17.1%	+4.1%

Table 2. CAPM estimates, full sample. The two largest alphas are gold (+8.5%, a very strong gold bull market) and technology (+4.1%); AGG's beta is zero and EFA/EEM show negative alpha. An optimiser fed these means will favour GLD and XLK. We treat that as a signal to be constrained, not trusted.

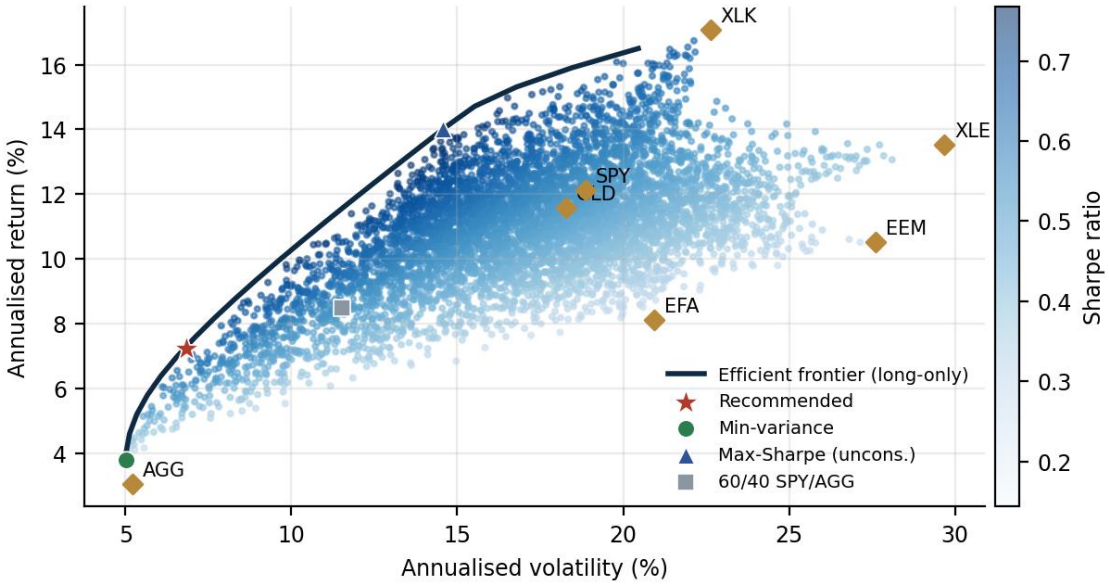


Figure 6. Mean-variance opportunity set: 6,000 random long-only portfolios, the efficient frontier, the seven assets and the candidate portfolios (in-sample).

Portfolio	SPY	EFA	EEM	AGG	GLD	XLE	XLK	Return*	Vol	Max DD
Recommended	5%	0%	0%	61%	19%	0%	15%	7.0%	6.8%	-17.9%
Min-variance	6%	0%	0%	92%	1%	1%	0%	3.7%	5.0%	-16.8%
Max-Sharpe (uncons.)	0%	0%	0%	4%	46%	0%	50%	13.6%	14.5%	-33.1%
Equal weight (1/7)	14%	14%	14%	14%	14%	14%	14%	9.8%	15.6%	-43.3%
60/40 SPY/AGG	60%	0%	0%	40%	0%	0%	0%	7.9%	11.3%	-35.8%

*Table 3. Candidate portfolios. *Return is the realised compound annual growth rate with monthly rebalancing.*

How we chose. The recommended portfolio solves: maximise Sharpe ratio subject to long-only weights, the IPS asset caps and an ex-ante volatility of at most 7%. The raw solution (SPY 5.0%, AGG 59.5%, GLD 19.2%, XLE 1.4%, XLK 15.0%) was rounded to whole percentages, with the 1.4% energy position dropped as immaterial and absorbed by AGG. Ex-ante it offers 7.2% return at 6.8% volatility. Against the alternatives: **minimum variance** (92% AGG) has the lowest volatility but a 3.7% return that fails the 6.5% objective; **maximum Sharpe** posts the highest Sharpe ratio (0.78) but is a two-asset bet on the in-sample winners with a 33% drawdown, far outside tolerance; **equal weight** and **60/40** breach the drawdown limit (43% and 36%). Only the constrained portfolio satisfies both the return objective and every risk limit.

6. Performance KPIs and robustness of the construction

KPI	Recommended	Min-variance	Max-Sharpe	Equal weight	60/40	SPY
CAGR	7.0%	3.7%	13.6%	9.8%	7.9%	10.9%
Annual volatility	6.8%	5.0%	14.5%	15.6%	11.3%	18.9%
Sharpe ratio	0.67	0.25	0.78	0.52	0.51	0.51
Sortino ratio	0.95	0.34	1.12	0.73	0.72	0.72
Beta vs SPY	0.22	0.07	0.56	0.76	0.59	1.00
Treynor ratio	0.21	0.19	0.20	0.11	0.10	0.10
Jensen alpha (annual)	+2.4%	+0.6%	+5.9%	+0.7%	+0.1%	-0.0%
Information ratio vs 60/40	-0.15	-0.44	0.57	0.33	n/a	0.48
Maximum drawdown	-17.9%	-16.8%	-33.1%	-43.3%	-35.8%	-55.2%
Calmar ratio	0.39	0.22	0.41	0.23	0.22	0.20

Table 4. Institutional KPIs, Jan 2005 to Oct 2026, monthly rebalancing, before transaction costs, risk-free rate 2.5%, CAPM market proxy SPY, information-ratio benchmark 60/40. Sortino uses the risk-free rate as minimum acceptable return.

Interpretation. Per unit of total risk the recommended portfolio earns a Sharpe ratio of 0.67, +31% relative to SPY and +32% relative to 60/40. Per unit of market risk the lead is larger: Treynor 0.21 against 0.10 for SPY, because beta is only 0.22. The Calmar ratio of 0.39 is the highest of the conservative candidates. The information ratio against 60/40 is -0.15: the portfolio does not try to beat 60/40 on return; it matches the return objective with 40% less volatility and half the drawdown.

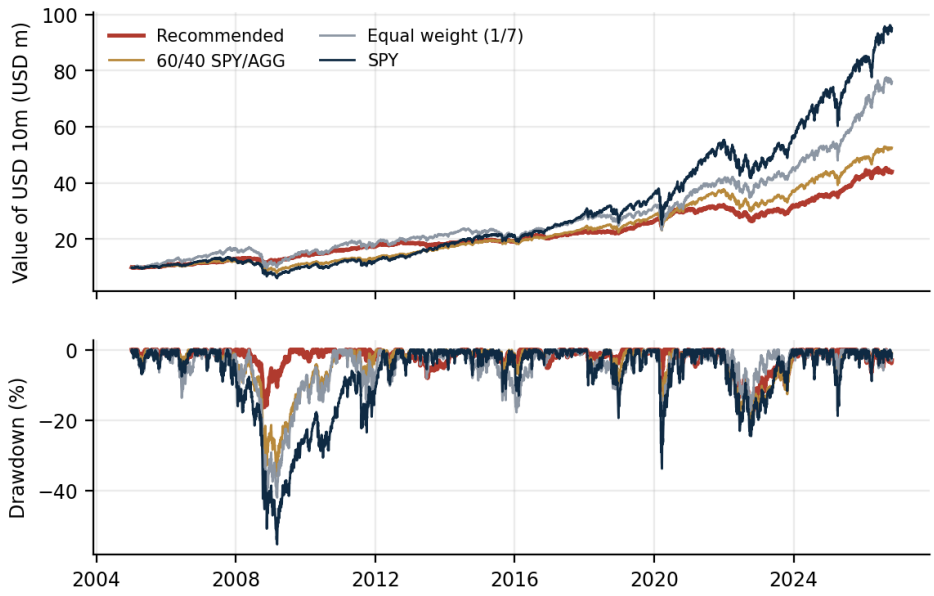


Figure 7. Growth of USD 10 million and drawdowns. The recommended portfolio (terminal values: Recommended USD 44.1m, 60/40 USD 52.6m, SPY USD 95.0m).

Return versus risk, honestly stated. Over 22 years SPY would have grown USD 10 million to USD 95m against USD 44m, but with a 55% drawdown that would have forced a spending-fund to sell near the 2009 low. The recommended portfolio's worst drawdown (17.9%, 2021-12 to 2022-10) came from the 2022 joint bond and equity decline, not from the 2008 crash; it was negative in 5 of 22 calendar years versus 3 for SPY, so it is not a loss-free strategy. Annual turnover from monthly rebalancing is 23% (two-sided), a drag of about 1.1 bp per year at 5 bp per trade, so transaction costs do not change any conclusion.

Out-of-sample test of the construction rule

Test period Jan 2015 - Oct 2026	Rule trained on 2005-14	60/40	Equal weight	SPY
CAGR	7.2%	9.0%	10.9%	13.7%
Annual volatility	7.0%	10.8%	13.7%	17.5%
Sharpe ratio	0.67	0.62	0.65	0.68
Maximum drawdown	-15.3%	-21.6%	-27.9%	-33.7%
Calmar ratio	0.47	0.41	0.39	0.41

Table 5. We re-ran the full procedure (same IPS, same 7% volatility cap) on 2005-2014 data only, which produced AGG 61%, GLD 20%, XLK 15%, XLE 4%, SPY 0%, and evaluated it untouched on 2015-2026. It held its risk profile (volatility 7.0% vs 6.8% predicted) and limited the drawdown to 15.3% against 21.6% for 60/40. Caveat: gold also rallied in 2005-2014, so this is not a fully independent test of the gold allocation.

7. Layers 2 and 3: Value-at-Risk and Expected Shortfall

We apply three required VaR methodologies to the same portfolio at 95% and 99% confidence, one-day horizon, USD 10 million. **Historical simulation** reads the empirical loss quantile of 5,488 days. **Parametric** assumes normal portfolio returns with sample mean and standard deviation (and, as a robustness variant, a Student-t fitted by maximum likelihood). **Monte Carlo** draws 400,000 scenarios from the seven-asset mean vector and covariance matrix, once from a multivariate normal and once from a multivariate Student-t with 3.6 degrees of freedom. ES is the average loss beyond VaR.

Method	95% VaR	95% ES	99% VaR	99% ES	99% VaR (USD)	99% ES (USD)

Historical	0.63%	0.97%	1.13%	1.62%	112,673	161,800
Parametric-Normal	0.68%	0.86%	0.97%	1.11%	96,867	111,384
Parametric-t	0.59%	0.93%	1.10%	1.60%	109,939	160,337
Monte Carlo (normal)	0.68%	0.86%	0.97%	1.12%	97,118	111,586
Monte Carlo (Student-t)	0.60%	0.96%	1.12%	1.62%	111,870	161,814

Table 6. One-day VaR and ES of the recommended portfolio (loss as % of NAV; USD on 10 million).

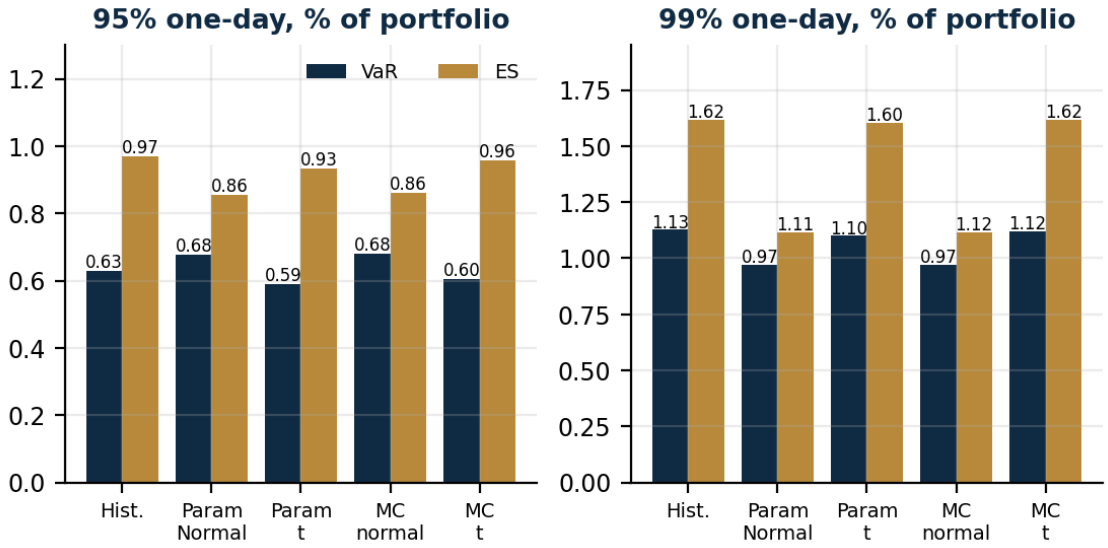


Figure 8. VaR and ES by method at 95% (left) and 99% (right).

Where the three methods disagree, and why

- **At 99%, normal understates risk.** Normal VaR is 0.97% versus 1.13% historical (14% lower) and normal ES is 31% lower. The normal distribution has thin tails, so the further into the tail we go the larger the error. The ES-to-VaR ratio is 1.44 historically but only 1.15 under normality.
- **At 95%, the ranking reverses.** Normal VaR (0.68%) exceeds historical (0.63%). With excess kurtosis near 18, most days are quieter than a normal curve with the same variance predicts, and the variance is inflated by a few extreme days. Normality is therefore too cautious in the body and too optimistic in the tail.
- **Monte Carlo is only as good as its inputs.** The normal Monte Carlo (0.97%) simply reproduces the parametric result (0.97%): sampling noise aside, simulation does not add information when the distributional assumption is the same. Switching the simulation to a Student-t reproduces the historical numbers (1.12% VaR, 1.62% ES), which tells us the heavy tail, not the method, is what matters.
- **Why report ES as well as VaR.** VaR is silent about losses beyond its threshold; two portfolios with the same 99% VaR can have very different worst cases. ES is coherent (sub-additive, so diversification can only reduce it), is what the Committee actually asks ("how bad could losses become?"), is the measure regulators adopted for market risk (FRTB uses 97.5% ES), and, as an average over many tail observations, is more informative about tail shape. Here the 99% ES is 44% above VaR, a gap the Committee should budget for.

8. Layer 4: Extreme Value Theory

Historical simulation says nothing about losses worse than those in the sample, and 99%+ quantiles rest on very few days. By the Pickands-Balkema-de Haan theorem, losses above a high threshold u follow a generalised Pareto distribution (GPD) with shape ξ and scale σ. We fit the GPD by maximum likelihood to the portfolio's daily losses (peaks over threshold). With the exceedance fraction F_u, VaR_c = u + (σ/ξ)[(F_u/(1-c))^ξ - 1] and ES_c = (VaR_c + σ - ξu)/(1-ξ), valid for ξ < 1.

Threshold selection (justified, not arbitrary)

We scanned thresholds from the 85th to the 98.5th percentile of daily losses and examined three diagnostics (Figure 9): the mean excess function, which should be roughly linear above a valid threshold; and the stability of the shape ξ and the modified scale σ* = σ - ξu, which should be flat. To avoid eyeballing, we formalised the rule: pick the threshold minimising the local standard deviation of ξ over a five-point window, with at least 75 exceedances. This selects the **91% quantile (u = 0.47% daily loss), 494 exceedances, ξ = 0.195, σ = 0.247%**. A positive shape confirms a heavy, power-law tail (tail index 1/ξ = 5.1).

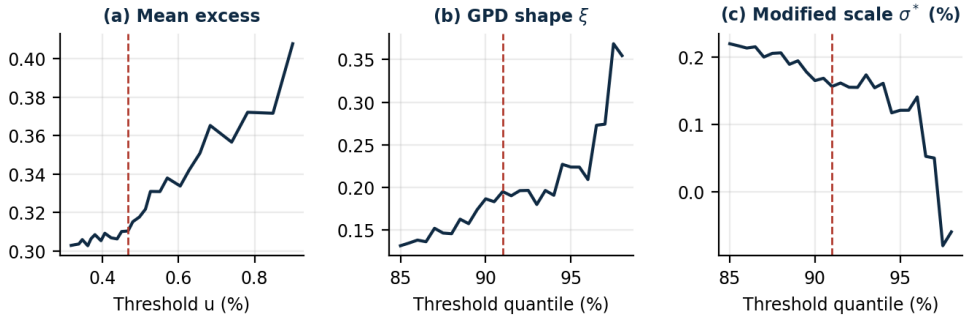


Figure 9. Threshold diagnostics; the dashed line marks the selected threshold.

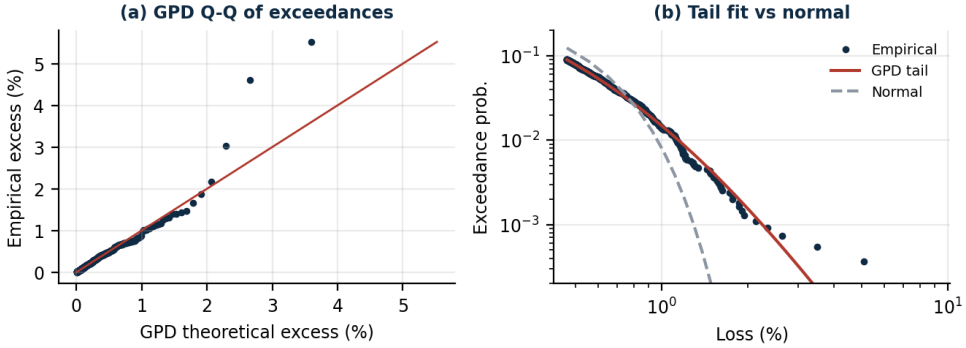


Figure 10. Goodness of fit. (a) Exceedances track the 45-degree line apart from the single worst day; (b) the GPD follows the empirical tail where the normal collapses. Kolmogorov-Smirnov p = 0.83; Cramer-von Mises parametric-bootstrap p = 0.45: no evidence against the GPD. Moving the threshold from the 90th to the 97th percentile shifts the 99% ES only between 1.614% and 1.618%.

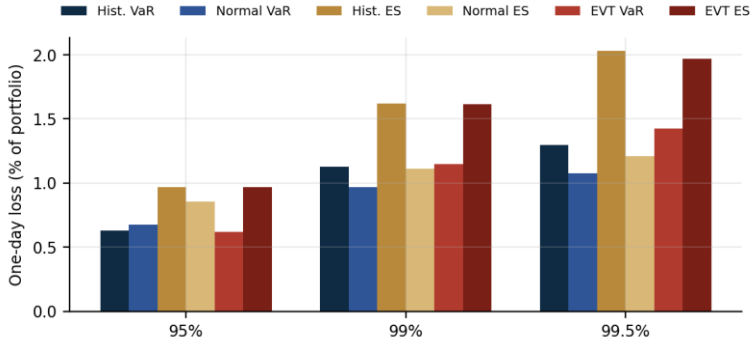


Figure 11. Historical, normal and EVT VaR and ES side by side at 95%, 99% and 99.5%.

Level	Normal VaR	Hist. VaR	EVT VaR	Normal ES	Hist. ES	EVT ES	EVT ES (USD)
95.0%	0.68%	0.63%	0.62%	0.86%	0.97%	0.97%	96,604
99.0%	0.97%	1.13%	1.15%	1.11%	1.62%	1.62%	161,769
99.5%	1.08%	1.29%	1.43%	1.21%	2.03%	1.97%	196,787
99.9%	1.30%	2.04%	2.25%	1.41%	3.62%	2.99%	298,921

Table 7. Tail estimates side by side. At 99% EVT and history agree (ES 1.62% for both) because 99% lies inside the data; the 90% bootstrap interval for the EVT 99% ES is 1.44% to 1.82%. Beyond the data, EVT is the only estimate with a statistical basis: at 99.9% it is 2.1 times the normal ES, and below the 3.62% historical figure, which rests on about 5 observations dominated by Oct 2008. Unconditional EVT assumes independent exceedances; Section 9 tests a volatility-filtered variant.

9. Layer 5: Model validation by out-of-sample backtesting

A risk model that has not been tested against reality is an opinion, not a measurement. For each trading day from 2007-12-27 to 2026-10-27 (4,738 forecasts, covering 2008, 2020 and 2022) we produced a one-day-ahead VaR using only the preceding 750 days (about three years), then recorded whether the next day's loss exceeded it. We compared five models: **Historical**; **Parametric-Normal**; **FHS-EWMA** (returns standardised by an EWMA volatility with decay 0.94, quantile taken from the standardised residuals, rescaled by today's volatility forecast); **EVT (unconditional)**, a GPD fit re-estimated every 10 days; and **EVT-EWMA (McNeil-Frey)**, the GPD fitted to standardised residuals. Tests: Kupiec proportion-of-failures (is the exception rate right?), Christoffersen independence (are exceptions unclustered?), their joint conditional-coverage test, and the Basel traffic light on rolling 250-day windows (green 0-4, yellow 5-9, red 10+ exceptions at 99%).

Model	99% exc. (exp. 47)	Kupiec p	Indep. p	Cond. cov. p	Last 250 (zone)	Basel G/Y/R windows	95% exc. (exp. 237)	95% Kupiec p	95% cond. cov. p
Historical	57	0.173	0.190	0.168	7 (Yellow)	85/7/8%	280	**0.005**	**<0.001**
Parametric-Normal	82	**<0.001**	**0.003**	**<0.001**	**11 (Red)**	66/23/11%	243	0.685	**<0.001**
FHS-EWMA	50	0.705	**0.017**	0.053	2 (Green)	88/12/0%	245	0.591	0.409
EVT (unconditional)	50	0.705	0.116	0.270	5 (Yellow)	85/7/8%	278	**0.008**	**<0.001**
EVT-EWMA (McNeil-Frey)	49	0.814	**0.015**	0.050	2 (Green)	88/12/0%	247	0.504	0.070

Table 8. Backtest results; p-values below 0.05 (reject) in red. 'Last 250' is the exception count in the most recent 250 forecasts.

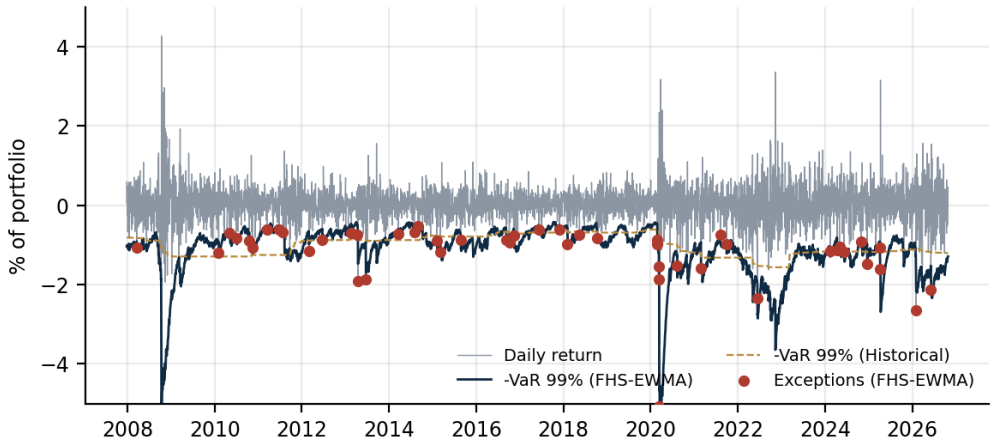


Figure 12. Out-of-sample 99% VaR from the selected FHS-EWMA model (navy) with exceptions (red), and the static historical VaR (dashed gold). The filtered VaR widens in 2008 and 2020 instead of lagging the crisis.

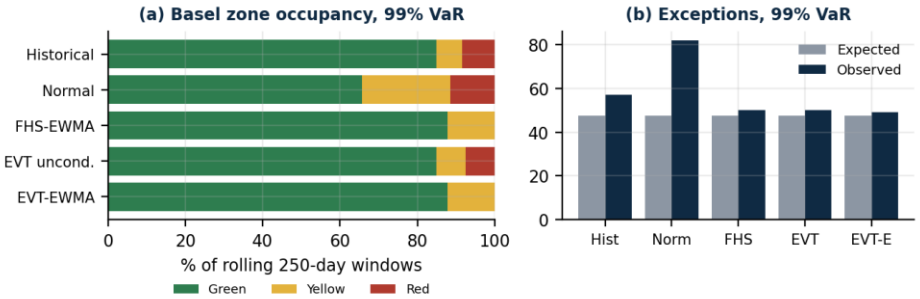


Figure 13. Basel traffic-light occupancy and exception counts, 99% VaR.

Findings. (1) **Parametric-Normal fails decisively**: 82 exceptions against 47 expected (Kupiec p <0.001), 11% of rolling windows in the red zone, and red today (11 exceptions in the last 250 days). This is the practical cost of ignoring fat tails. (2) **Unfiltered models cluster:** Historical and unconditional EVT get the 99% frequency about right but fail at 95% and spend 8% and 8% of windows in the red zone, because exceptions bunch in 2008 (12) and 2020 (13). (3) **FHS-EWMA is our**

operating model: 50 99% exceptions (expected 47), Kupiec p = 0.705, no red-zone window at any time, green zone today (2 exceptions), and it passes all three tests at 95% (p = 0.591, 0.221, 0.409). Its weakness: the 99% independence test is rejected (p = 0.017; McNeil-Frey p = 0.015), so some clustering remains. No model passes every test at both levels; we use FHS-EWMA for daily limits and EVT beyond 99%.

10. Stress testing: historical crises and hypothetical shocks

Each scenario applies observed or assumed returns to the weights held at the start of the window (buy-and-hold, no rebalancing, no costs). Historical windows are the peak-to-trough episodes the brief names; the two hypothetical scenarios were designed by the team to attack the portfolio where it is weakest, since a stress test that only replays the past tests what already happened.

Scenario	Recommended	USD impact	60/40	Equal wt	SPY	Worst in-window DD (Rec.)
GFC (1 Sep 2008 - 9 Mar 2009)	-5.3%	-0.53m	-27.0%	-31.4%	-46.4%	-12.7%
COVID crash (19 Feb - 23 Mar 2020)	-7.6%	-0.76m	-20.6%	-26.6%	-33.4%	-11.1%
2022 rate-hiking (3 Jan - 14 Oct 2022)	-17.6%	-1.76m	-20.6%	-12.8%	-23.8%	-17.6%
H1 Real-rate shock (+200bp)	-14.7%	-1.47m	-15.6%	-17.1%	-18.0%	n/a
H2 Stagflation / oil shock	-4.0%	-0.40m	-9.2%	-4.1%	-12.0%	n/a

Table 9. Portfolio returns under each scenario (USD impact on 10 million).

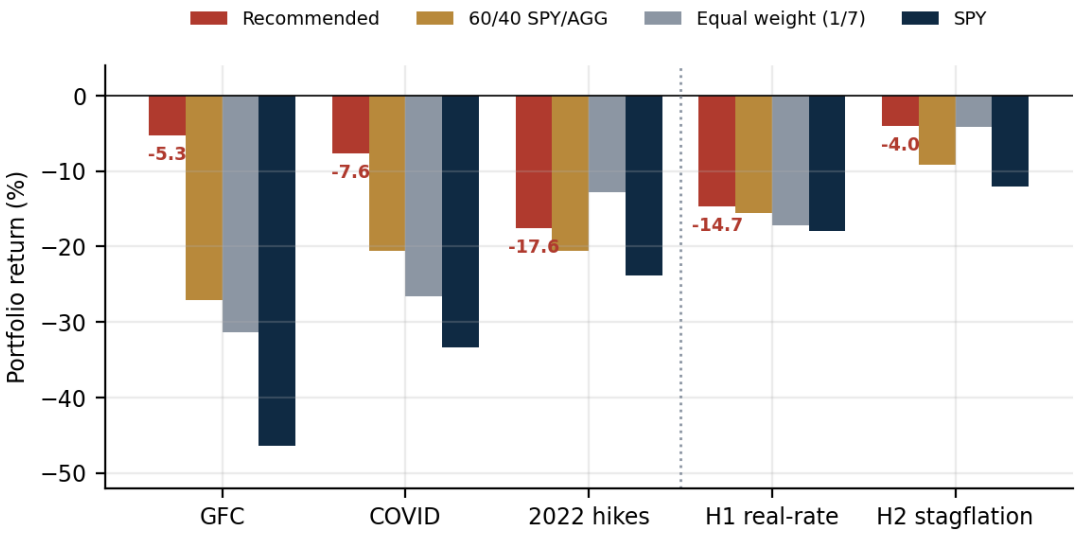


Figure 14. Scenario returns for the recommended portfolio against three comparators (left of the dotted line: historical; right: hypothetical).

Asset returns in scenario	SPY	EFA	EEM	AGG	GLD	XLE	XLK
GFC	-46%	-49%	-48%	+2%	+11%	-48%	-41%
COVID crash	-34%	-33%	-31%	-1%	-4%	-56%	-31%
2022 rate-hiking	-24%	-28%	-30%	-15%	-9%	+45%	-33%
H1 shock	-18%	-20%	-25%	-12%	-12%	-5%	-28%
H2 shock	-12%	-14%	-18%	-5%	+10%	+25%	-15%

Table 10. Asset-level returns behind each scenario. H1 is a +200bp real-rate shock (AGG at roughly six years of duration loses 12%, long-duration technology falls 28%, gold falls 12% as real yields rise). H2 is a stagflationary oil shock (XLE +25%, GLD +10%, equities and bonds down).

Reading the results

- **Equity crashes: the design works.** In the GFC window AGG (+1.9%) and GLD (+10.8%) rose while equities lost about 45%, so the portfolio lost 5.3% against 46% for SPY. In the COVID crash it lost 7.6% against 33%.

- **Inflation and rate shocks: the design is exposed.** In 2022 AGG fell 15.1%, GLD 9.1% and XLK 33% together, a 17.6% loss that is the portfolio's worst drawdown ever. Our hypothetical H1 reproduces and extends this mechanism (-14.7%, USD 1.47m). All scenario losses remain inside the 20% tolerance, but 2022 used 88% of it. An energy overlay would have helped there (equal weight, which holds 14% XLE, lost only 12.8%).
- **We tested an energy hedge and did not adopt it.** Replacing 4% of AGG with XLE improves 2022 to -15.0% and H2 to -2.8%, but worsens the GFC (-7.3% vs -5.3%), the COVID crash (-9.8% vs -7.6%) and 99% ES (1.75% vs 1.62%). Hedging the last war costs protection in the next equity crash; we list it as a Committee option, not a recommendation.
- **Volatility stress.** Doubling every asset's volatility while preserving means and correlations lifts the portfolio's one-day 99% ES to 3.28% (USD 328k) and its maximum drawdown to 38%, which would breach the 20% tolerance. The limits are sized for a 7% volatility portfolio, not a doubled one, which is why we propose the drawdown trigger in Section 12.

11. From metrics to decisions: risk budgeting, regimes and trade-offs

Risk budgeting and position sizing

Capital weights say little about risk. Using each position's marginal contribution, we decompose portfolio volatility (covariance-based) and historical 95% Expected Shortfall (the average return of each asset on the days the portfolio lost more than its 95% VaR; 0.97% total).

Asset	Capital weight	Share of volatility	Share of ES (95%)	Marginal ES per unit weight	Risk / capital
SPY	5%	9%	9%	1.72%	1.8x
AGG	61%	26%	23%	0.37%	0.4x
GLD	19%	33%	35%	1.76%	1.8x
XLK	15%	32%	33%	2.16%	2.2x

Table 11. Risk budget of the recommended portfolio (EFA, EEM and XLE hold zero weight).

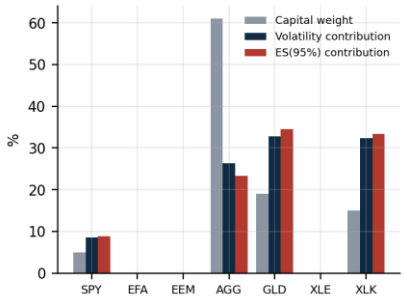


Figure 15. Capital versus risk contribution.

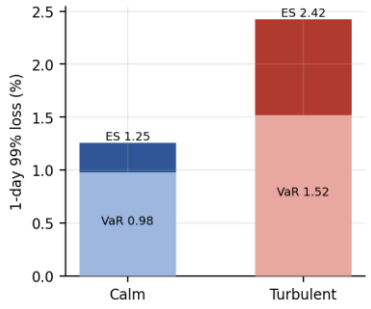


Figure 16. 99% VaR and ES by volatility regime.

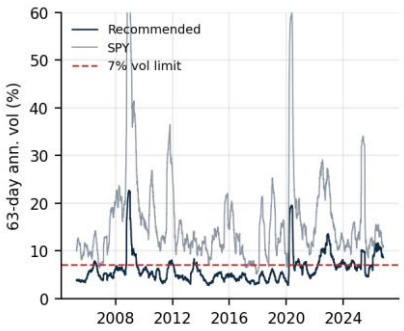


Figure 17. Rolling volatility versus the 7% limit.

Position-sizing rules that follow. (1) AGG is 61% of capital but only 26% of volatility, so it is genuinely the capital-preservation sleeve. (2) GLD (35% of ES) and XLK (33%) each carry about 1.8 and 2.2 times their capital share in risk; the IPS caps them at 20% and 15% and a 35% risk-contribution ceiling binds GLD tightly. Any increase in either requires a corresponding reduction in the other or a lower overall limit. (3) SPY is the cheapest equity risk per unit of marginal ES (1.7% versus 2.2% for XLK) and is the first place to add equity if the Committee later raises its risk appetite.

Regime-conditioned risk (extension)

We split days by the prior-day trailing 63-day volatility of SPY (a proxy for a volatility regime that uses no look-ahead): turbulent means the top quartile (above 18.3% annualised). In turbulent regimes the portfolio's annualised volatility is 9.6% against 5.6% in calm regimes, and its one-day 99% ES is **2.42% versus 1.25%** (USD 242k versus USD 125k). The single unconditional ES (1.62%) therefore hides a 1.9x swing: the 2.0% ES limit holds in calm regimes but would be breached in turbulent ones. Rolling

63-day volatility of the portfolio exceeded the 7% target on 25% of days (peak 23%), so the 7% should be read as a through-the-cycle target, with the daily risk model (FHS) used for day-to-day limits. SPY's 63-day volatility is currently 10.7% (calm regime).

Quantifying the trade-off between return and risk reduction

Volatility target	Max return (long-only)	99% VaR	99% ES	ES in USD	Added ES per +1% return
6%	6.3%	0.96%	1.44%	144,398	-
7%	7.5%	1.14%	1.65%	164,542	0.17%
8%	8.5%	1.31%	1.89%	189,320	0.25%
10%	10.3%	1.67%	2.38%	238,489	0.27%
12%	11.9%	2.02%	2.86%	285,942	0.28%
14%	13.6%	2.41%	3.33%	333,071	0.29%

Table 12. Efficient-frontier trade-off (unconstrained long-only, in-sample). Each additional 1 percentage point of expected return costs roughly 0.25 percentage points of one-day 99% ES, about USD 25,341 per day on USD 10 million. The IPS-constrained recommendation sits at the 7% row: return 7.2% versus 7.5% for the unconstrained optimum, so the diversification and concentration rules cost about 0.2 percentage point of return.

12. Investment committee memorandum

To Investment Committee, The Serendib University Endowment Fund
From Team Quantix, quantitative research division
Date 30 October 2026
Re Recommendation on the proposed USD 10 million mandate

Recommendation: ACCEPT the mandate as proposed, with the portfolio below and the risk framework in the table. We do not recommend rejecting it (a conservative objective is achievable with measured risk) and do not recommend loosening any limit.

Portfolio: AGG 61%, GLD 19%, XLK 15%, SPY 5%; monthly rebalancing; long-only, unleveraged. Expected return about 7.2% a year at about 6.8% volatility.

Does the portfolio meet the Committee's own tests?

Limit	Requirement	Portfolio	Status
Return objective	At least 6.5% p.a.	CAGR 7.0%; out-of-sample rule 7.2%	**Met**
Volatility	At most 7%	6.8% full sample (rolling 63-day above 7% on 25% of days)	**Met on average**
Tail risk	1-day 99% ES at most 2.0% (USD 200k)	1.62% (EVT 1.62%); turbulent regime 2.42%	**Met; breached in turbulent regime**
Drawdown	At most 20%	17.9% (Dec 2021 to Oct 2022)	**Met, 2.1 pts headroom**
Stress loss	At most 20% in any scenario	Worst: 2022 -17.6%; H1 -14.7%	**Met**
Concentration	No asset above 35% of risk	GLD 35% of ES, XLK 33%	**Met, binding**
Model validity	No red-zone VaR model	FHS-EWMA green (2 exceptions in 250 days)	**Met**

The three questions the Committee will ask

What can go wrong? A simultaneous fall in stocks, bonds and gold, as in 2022 (-17.6%) or our +200bp real-rate scenario (-14.7%); a gold reversal (GLD fell 10.3% in one day on 2026-01-30, a 2.6% loss for the portfolio); a prolonged positive stock-bond correlation (now +0.45). **How bad could losses become?** On the worst 1% of days the average one-day loss is about USD 162k (ES); at the 1-in-1,000-day level, USD 299k (EVT); over a crisis, 5% to 18% of capital (USD 0.5m to 1.8m) on past evidence, and in a volatility-doubling shock 38%. **How confident are we?** Highly confident in the 95-99% daily numbers (validated on 4,738 out-of-sample days; the 99% ES 90% confidence band is 1.44% to 1.82%); moderately confident in the 99.9% figure (it extrapolates, shape uncertainty); least confident in the expected-return assumption, which relies on a 22-year sample with a historic gold rally.

Proposed monitoring and action triggers

- **Daily:** FHS-EWMA 99% ES against the 2.0% limit (about USD 200k); a Basel yellow zone (5+ exceptions in 250 days) triggers a model and exposure review. In turbulent regimes (SPY 63-day volatility above 18%) the Committee should expect ES near 2.4% and pre-agree a hard cap of 2.5%.
- **Drawdown trigger:** at -12% from the high-water mark, convene the Committee and review the GLD and XLK risk contributions; at -17% (3 points before the tolerance), move the equity-like sleeves (XLK, SPY) toward AGG as pre-authorised.
- **Correlation watch:** if the 126-day SPY-AGG correlation stays above +0.5 for 60 trading days (it is +0.45 now and touched 0.60 in Sep 2026), re-run the construction with the current covariance and consider the energy overlay tested in Section 10.
- **Annual:** re-estimate means and covariances, re-run the full backtest, and re-confirm the IPS with the Committee.

Decision options considered. Accept as proposed (recommended); adjust the risk limit (not needed, but the ES limit should be explicitly regime-aware); reject (not warranted, since the objective is achievable within the stated limits).

13. Assumptions and limitations, AI disclosure and reproducibility

Assumptions and limitations: what the model does not capture

- **The past is the only input.** All estimates use 2005-2026 history. That sample contains long gold bull runs, technology's long outperformance and an unusually long period of falling then low interest rates. Realised means (especially GLD's +8.5% CAPM alpha and XLK's) are unlikely to repeat in full; the optimiser's tilt to them is the main model-risk in the recommendation. Expected returns carry far more estimation error than risks.
- **The weights are chosen with hindsight.** The recommended weights use the full sample, and the backtest holds them fixed. The VaR models are genuinely out-of-sample (rolling windows), but the portfolio is not. The 2005-14 re-estimation in Section 6 mitigates but does not remove this, since gold also performed strongly in that period.
- **Risk-free rate and benchmarks.** We used a constant 2.5% risk-free rate in place of the FRED 3-month Treasury series (the approved source) because the analysis environment had no live data access; the series should be substituted in the repository. Sharpe, Sortino, Treynor and alpha shift modestly with this input; VaR, ES, EVT and stress results do not depend on it.
- **Distributional and dynamic assumptions.** Historical VaR assumes the sample is representative; normal VaR assumes thin tails and constant variance (rejected); EVT assumes independent exceedances above a constant threshold (volatility clustering violates this) and extrapolates beyond the data, so 99.9% estimates carry wide uncertainty. FHS-EWMA uses a fixed decay of 0.94 and a zero-mean assumption; its 99% independence test is rejected, meaning some exception clustering remains.
- **One-day horizon.** VaR and ES are one-day measures. We did not scale them to 10-day or longer horizons; liquidity-adjusted multi-day losses in a crisis could be larger than square-root-of-time scaling suggests.
- **Correlations and regimes change.** Diversification relies on low correlation between AGG, GLD and equities, which failed in 2022 and is weak now. Our regime split is a simple volatility-quantile rule, not a fitted regime-switching model, and the hypothetical scenarios are our judgement, not forecasts.
- **Implementation is idealised.** ETF prices are assumed executable at the close with fractional shares, monthly rebalancing and no taxes, bid-ask spread, market impact, ETF tracking error, tracking-difference between adjusted prices and true total returns, currency effects beyond USD pricing, or fee drag. Costs of about 1 bp a year at 5 bp per trade were estimated, not simulated in the KPIs.

- **What is not modelled at all.** Inflation-linked liabilities, the Endowment's spending policy and liquidity needs, credit and default risk inside AGG, counterparty and operational risk, the possibility that an ETF is closed or suspended, and macro scenarios beyond the five tested. The model measures market risk of a long-only ETF portfolio, nothing broader.

Use of AI and coding assistants (disclosure required by Section 8.7 of the brief)

Item	Disclosure
AI tool(s) used	Claude (Anthropic).
Purpose	Code scaffolding for the analysis pipeline and charts, executing the quantitative workflow on the supplied price file, and drafting and formatting of this report.
Human validation	Risk calculations (historical and normal VaR/ES, GPD fit) were cross-checked against the team's QuantEdge toolkit, which has its own unit-test suite, and agree to within 0.002 percentage points. [Team to confirm before submission: who reviewed the assumptions, the IPS limits and the recommendation, and what was changed.]
Material AI-generated components	Analysis code, figures and the first draft of the text. Portfolio construction choices, risk interpretation, stress-scenario design and the final recommendation must be understood and defended by the team in the live pitch.

Reproducibility

Every statistic and chart in this report is produced by the delivered analysis script (analysis.py) from the single file price_data.xlsx, with fixed random seeds (Monte Carlo 42, bootstrap 7 and 11); the report document is then built from its results file. The QuantEdge toolkit supplied with the team files (data validation, optimisation, performance, VaR/ES, EVT, backtesting, stress testing; CLI entry point app.py) was used as an independent cross-check. The Streamlit interface required by Section 6 of the brief is a separate deliverable and is not described here. No proprietary data were used; the source is Yahoo Finance adjusted closes, and FRED should supply the risk-free rate (see above).